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Lesson 40 Waves

海浪 状语从句 定语从句

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signature / sɪgnətʃə(r) /
ray / reɪ /
rhythm / 'rɪðəm /
exquisite / ɪk'skwɪzɪt /
crest / krest /
vertical / 'vɜːtɪk(ə)l /
actuality / ˌæktʃu'æləti /
particle / 'pɑːtɪk(ə)l /
undulate / 'ʌndjuleɪt /
gravitational / ˌgrævɪ'teɪʃən(ə)l /
infinity / ɪn'fɪnəti /
energize / 'enədʒaɪz /
transmit / trænz'mɪt /
phenomena / fə'nɒmɪnə /
trough / trɒf /
horizontal / ˌhɒrɪ'zɒnt(ə)l /
catastrophic / ˌkætə'strɒfɪk /
maturity / mə'tʃʊərəti /
tremor / 'tremə(r) /

海浪的诞生 (笔记最后修改于:2026-04-08 12:59:18)

※ Waves are the children of the struggle between ocean and atmosphere, the ongoing signatures of infinity.

『 **child of...** ...的孩子/产物 』

- The atomic bomb is **the child of** 20th century scientific developments.
- The romantic movement was **the child of** the Great Revolution.
- Innovation is often the child of necessity. 创新往往是必需品的产物。
- These intricate designs are the children of a long tradition of craftsmanship. 这些复杂的设计是长期工艺传统的产物。

『 **the struggle between... and...** ...与...之间的斗争/抗衡 』

- **the struggle between** good **and** evil 善与恶之间的斗争

- History often records **the struggle between** progress **and** tradition.历史经常记录进步与传统之间的斗争。

『**ongoing** adj. continuing to exist or develop 持续（存在或发展）的』

- The police investigation is **ongoing**.
- Training is part of our **ongoing** career development program.
- The starry night sky can be seen as **the ongoing signatures of infinity**.星空夜景可以被视为无限的持续标记。
- The continuous flow of the river represents **the ongoing signatures of infinity**.河流的持续流动代表着无限的持续标记。

an ongoing debate 一场持续进行的辩论	There is **an ongoing debate** among scientists about the long-term effects of this new technology. 科学家们正就这项新技术的长期影响进行着一场持续的辩论。
an ongoing discussion 一次正在进行的讨论	We have **an ongoing discussion** with our partners regarding the budget for next year. 我们正与合作伙伴就明年的预算进行着持续的讨论。
an ongoing process 一个持续进行的过程	Learning a new language is **an ongoing process** that requires constant practice and patience. 学习一门新语言是一个持续的过程，需要不断的练习和耐心。

※ Rays from the sun excite and energize the atmosphere of the earth, awakening it to flow, to movement, to rhythm, to life.

『**ray** n. [C] a straight narrow beam of light from the sun or moon 光线』

- The room darkened as a cloud hid the sun's **rays**.
- Plants convert energy from the **rays from the sun** into food.植物将来自太阳的光线的能量转化为食物。

X-rays X射线；X光	The doctor ordered **X-rays** to check if any bones were broken after the fall. 医生要求拍X光片，以检查摔倒后是否有骨折。
ultraviolet rays (UV rays) 紫外线	Sunscreen protects your skin from the harmful effects of **ultraviolet rays**. 防晒霜可以保护你的皮肤免受紫外线的有害影响。

『**energize** v. to make someone feel full of energy or enthusiasm 使某人充满能量或热情』

- a refreshing and **energizing** fruit drink 一种清新提神的水果饮料
- The discovery of the new particle helped to **excite and energize** the entire research team.新粒子的发现帮助激发并为整个研究团队注入能量。

『**awaken sb. to sth.** cause sb. to become aware of sth. 使某人意识到某事』

- **awaken** society **to** the dangers of drugs
- During the 1930s Churchill tried to **awaken** the British people **to** the need for rearmament.
- It has **awakened** millions **to** political consciousness and recognition of the danger of fascism.
- The spring rain seems to **awaken** the dormant seeds **to life**.春雨似乎唤醒了休眠的种子以使其获得生命。
- The new policy is intended to **awaken** the market **to greater competition**.这项新政策旨在唤醒市场以迎接更大的竞争。

『 **rhythm** n. [U and C] 节奏 』

- the steady **rhythm of** her heartbeat
- She started moving to the **rhythm of** the music.

※ The wind then speaks the message of the sun to the sea and the sea transmits it on through waves -- an ancient, exquisite, powerful message.

『 **speak** v. to say or state sth. 说出来 』

- She was clearly **speaking** the truth.
- He **spoke** the final words of the play.
- He **spoke** only two words the whole evening.
- The poet's verses **speak the message of hope to** a generation seeking direction.诗人的诗句向寻求方向的一代人传达了希望的信息。

『 **transmit sth. on through...** 通过...传递/发送某物 』

- The radio station **transmits the signal on through** a network of local towers.该电台通过本地塔台网络发送信号。
- Bacteria can **transmit resistance on through** horizontal gene transfer.细菌可以通过水平基因转移传递耐药性。

波浪的特征 (笔记最后修改于:2026-04-08 13:17:35) 3

※ These ocean waves are among the earth's most complicated natural phenomena.

『 **be among ...** 是...之一；包括在...之中 』

- He **was among** the last to leave.
- **Among** those present **were** the Prime Minister and her husband. 表系主（引起了倒装）
- **Among** them will be Debbie's mother, who swam the Channel herself when she was a girl.
- Designing a secure system is **among the most complicated** tasks in software engineering.设计一个安全的系统是软件工程中最复杂的任务之一。

- The structure of the human brain is **among the most complicated** biological systems known.人类大脑的结构属于已知最复杂的生物系统之一。

※ The basic features include a crest (the highest point of the wave), a trough (the lowest point), a height (the vertical distance from the trough to the crest), a wave length (the horizontal distance between two wave crests), and a period (which is the time it takes a wave crest to travel one wave length).

『 **crest** n. the top or highest point of something such as a hill or a wave 山峰，浪尖 』

- We reached the **crest** of the hill just as the sun was setting.我们爬到山顶时，太阳正好落下。
- The **crest** of the blue heron is particularly prominent during the mating season. 大蓝鹭的羽冠在繁殖季节尤为明显。
- The family **crest** was engraved on the silver platter. 家族徽章刻在银盘上。

crest of ... ...的顶端；...的巅峰	• It took us over an hour to reach the **crest** of the hill. • surfers riding the **crest** of the wave • Surfers waited for hours to catch the perfect **crest of** a wave. 冲浪者等了好几个小时，只为抓到一个完美的浪尖。 • She is currently on the **crest of** a wave of popularity following her latest film. 随着最新电影的上映，她正处于人气巅峰。

『 **trough** n. the hollow area between two waves 波谷/低谷 』

- The economy is currently in a deep **trough**, but analysts predict a recovery soon.经济目前处于一个深谷中，但分析师预测很快会复苏。
- Waves are measured from the top (**crest**) to the bottom (**trough**). 海浪是从波峰到波谷进行测量的。
- From peak to **trough** more than 8m jobs were lost. 从峰值跌到低谷造成800万份工作流失。

> ..., and a period (which is the time ~~(which)~~ it takes a wave crest **to travel one wave length**). 补全

波浪的描述 (笔记最后修改于:2026-04-09 05:03:16) 6

※ Although an ocean wave gives the impression of a wall of water moving in your direction, in actuality waves move through the water leaving the water about where it was.

『 **give (sb.) the impression of...** 给人一种...的印象 』

- He **gives the impression that** he really doesn't care.

- The sudden movement of the door made it sway slightly and it **gave me the impression** that it was about to leap out at me.
- The new software update **gives the impression of** being much faster than the old version.新的软件更新给人一种比旧版本快得多的印象。
- Despite her confidence, her trembling hands **gave the impression of** deep nervousness.尽管她很自信，但她颤抖的双手给人一种深度紧张的印象。

『 **be under the impression (that) ...** to believe that sth. is true when it is not 误以为... 』

- I'm sorry, I **was under the impression that** you were the manager.
- Though the crew **were** at first **under the impression that** the lost ship had been found, the contents of the sea-chest proved them wrong.

『 **wall** n. thing similar to a wall in its appearance or effect （样子或作用）似墙之物 』

- Firefighters battled **the walls of flames**.
- The investigators were confronted by **a wall of silence**.
- The tidal wave formed **a terrifying wall of water**.
- The Titanic turned just in time, narrowly missing **the immense wall of ice** which rose over 100 feet out of the water beside her.

wall of fire 火墙；烈焰屏障	The firefighters were suddenly confronted by a massive **wall of fire** as the wind shifted. 风向转变时，消防员们突然面对一面巨大的火墙。
wall of water 水墙；巨浪（如海啸或洪水）	Witnesses described the tsunami as a terrifying **wall of water** rushing toward the shore. 目击者将海啸描述为冲向岸边的恐怖水墙。
wall of noise / sound 震耳欲聋的声音；声音障壁	As the band walked onto the stage, they were met with a **wall of noise** from the cheering crowd. 当乐队走上舞台时，迎接他们的是人群发出的排山倒海般的欢呼声。

『 **of** 做比喻（前面是喻体，后面是本体） 』

- Jane is **a submissive lamb of a girl**.
- There is always hope that in **its labyrinth of musty**, dark, disordered rooms a real rarity will be found amongst the piles of assorted junk that litter the floors.

a boy of a girl 假小子一样的姑娘	She was a real **boy of a girl**, always climbing trees and playing football with her brothers. 她是个十足的假小子，总是和哥哥们一起爬树、踢足球。
an angel of a wife 天使般的妻子	He knew he was lucky to have such **an angel of a wife** who supported him through the hardest times. 他知道自己很幸运，能有这样一位在最艰难时期支持他的天使般的妻子。

a palace of a house 宫殿般的房子；豪宅	After winning the lottery, they moved into **a palace of a house** with twelve bedrooms and a private lake. 中彩票后，他们搬进了一栋像宫殿一样的豪宅，里面有十二间卧室和一个私人湖泊。
a garden of Switzerland 花园般的瑞士	With its pristine lakes and lush green mountains, people often describe the country as **a garden of Switzerland**. 凭借其纯净的湖泊和郁郁葱葱的青山，人们常将这个国家描述为花园般的瑞士。

『 **in sb.'s direction** 朝某人的方向；向某人（看、走等） 』

- I waved, but she didn't even look **in my direction**. 我挥了挥手，但她甚至都没朝我这边看一眼。
- Fortunately for me, as I was wondering what to do next, there appeared on the horizon a man on horseback, riding **in my direction**.

in the direction of ... 朝...的方向	• Tom went off **in the direction of** the post office. • They headed off **in the direction of** the station. 他们朝车站方向走去。 • The wind is blowing **in the direction of** the coast. 风正往海岸方向吹。
in every direction 朝四面八方； **in all directions** 向各个方向	• It is a harmless habit, but it litters up her desk to such an extent that every time she opens it, loose bits of paper fall out **in every direction**. • The elephant is not immediately susceptible to such blandishments, however, and usually lashes fiercely with its trunk **in all directions**. • The crowd scattered **in all directions** when the rain started. 下雨时，人群向四面八方散开。 • From the top of the hill, the woods stretched away **in every direction**. 从山顶望去，森林向四面八方延伸开来。

『 **in actuality** = in reality / actually 事实上;实际上 』

- **In actuality**, it's much more complex than that.
- The woman he had seen performing on stage was **in actuality** quite older than she had seemed.
- The plan seemed simple, but **in actuality**, it was incredibly complex to execute. 这个计划看起来很简单，但实际上执行起来却极其复杂。
- He claimed to be a billionaire, but **in actuality**, he was deeply in debt. 他声称自己是亿万富翁，但实际上他负债累累。

※ If the water was moving with the wave, the ocean and everything on it would be racing in to the shore with obviously catastrophic results.

『 **were ⇒ was** 在当代英语中的趋势（虚拟语气） 』

- If she **was** here, she could speak on my behalf.
- If I **was** rich, I would buy you anything.
- If it **was** to rain, the ropes would snap.
- if I **were** you ⇒ **were** I you 语言习惯，不能改
- as it **were** 固定搭配
- if it **were** not for ⇒ **were** it not for 固定搭配

波浪的存在形式 (笔记最后修改于:2026-04-09 03:21:30) 1

※ An ocean wave passing through deep water causes a particle on the surface to move in a roughly circular orbit, drawing the particle first towards the advancing wave, then up into the wave, then forward with it and then -- as the wave leaves the particles behind -- back to its starting point again.

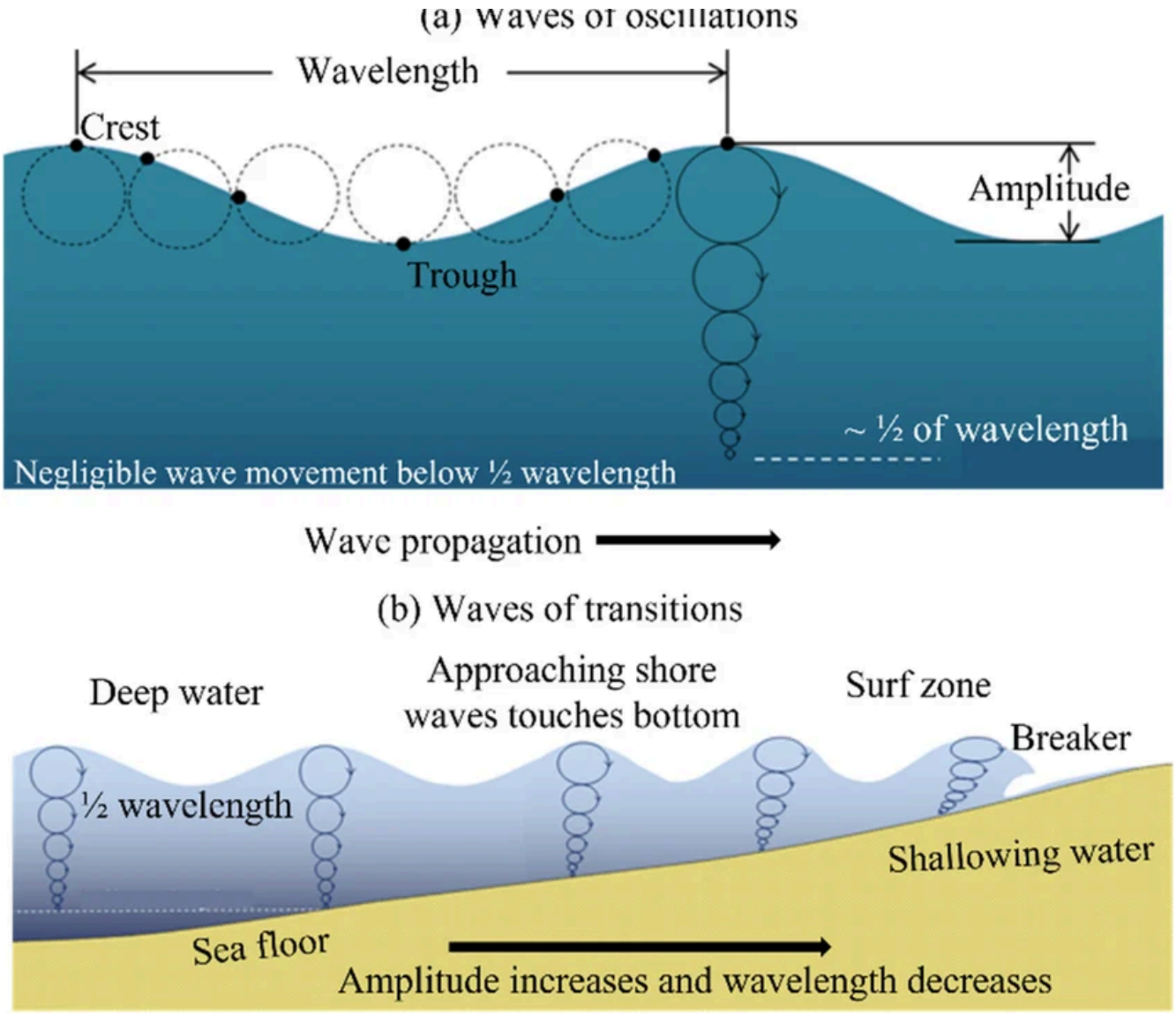
『 **particle** n. [C] a very small piece of sth. 微粒 』

- There's not a **particle** of truth in what he said. 他所说的话中没有一丝事实。
- Have you ever gotten a piece of sand or dirt or other small **particle** in your eye? 你曾经有过沙子、灰尘或其他小颗粒进入眼睛的经历吗?
- The energy of a light photon or **particle** is different for each color. 可见光子或粒子的能量因颜色不同而不同。

dust particles 尘埃微粒；粉尘	You can see **dust particles** dancing in the sunlight streaming through the window. 你可以看到尘埃微粒在透过窗户射进来的阳光中飞舞。
tiny particles of soil 极小的土壤颗粒	Plants absorb nutrients that are attached to **tiny particles of soil**. 植物吸收附着在极小土壤颗粒上的养分。

『 **passing through deep water** 穿过深水区 』

- The submarine was detected while **passing through deep water** far from the coast. 这艘潜艇在远离海岸的深水区穿行时被侦测到。
- The energy of the storm remained intact while **passing through deep water**. 风暴的能量在穿过深水区时保持完整。



海浪从成熟到消亡过程 (笔记最后修改于:2026-04-09 04:19:45)

※ From both maturity to death, a wave is subject to the same laws as any other 'living' thing.

> From **both** maturity to death 这里的 **both** 是一个强调词（Intensifier），用来强调范围的两端，表示“从……到……这两者过程中的全部”。
> 通常我们学的是 both... and...（两者都），但在 from both A to B 这个结构里，both 的作用是预示并强调接下来的两个端点（maturity 和 death）。
> 这种用法在文学或正式修辞中较多，目的是为了表达一种“完整性”。你可以把它理解为：“从成熟直到死亡，这两端及其间的一切。”

- **From** maturity **to** death...（从成熟到死亡……）—— 只是陈述一个时间段。
- From **both** maturity to death... —— 强调无论是成熟期还是死亡期（以及中间的过程），这个规律都适用。它把“成熟”和“死亡”作为两个对立但并列的点提出来加强语气。
- The company went through several phases **from both maturity to death** after the market changed.市场变化后，这家公司经历了从成熟到消亡的几个阶段。

- The life cycle of a star extends **from both maturity to death** over billions of years.恒星的生命周期从成熟到消亡绵延了数十亿年。

『 **be subject to ...** 1. if you are subject to a rule, law, penalty etc, you must obey the rule or pay an amount of money 必须服从或支付 』

- As a diplomat, he **is not subject to** local laws.
- Violators **are subject to** a $100 fine. Cars are subject to a high domestic tax.
- All citizens **are subject to the same laws** and regulations, regardless of their status.所有公民，无论其地位如何，都受制于相同的法律和法规。
- Even microscopic life forms **are subject to the same laws** of thermodynamics.即使是微小的生命形式，也受制于同样的热力学规律。

『 **be subject to ...** 2. often having, suffering or undergoing sth 常有、常患或常遭某事物 』

- **Are** you **subject to** colds?
- Trains **are subject to** delays after the heavy snowfalls.
- The timetable **is subject to** alteration.

『 **be subject to ...** 3. depending on sth. as a condition 需要某事物（为先决条件），取决于... 』

- The plan **is subject to** the director's approval.
- Goods will be sent out within 14 days, **subject to** availability.

> So let us begin anew—remembering on both sides that civility is not a sign of weakness, and sincerity **is always subject to** proof. Let us never negotiate out of fear, but let us never fear to negotiate. — John F. Kennedy

『 **as 引导比较状语从句（省略be）** 』

- From both maturity to death, a wave is subject to the same laws as any other 'living' thing~~ (is)~~. 主语里有be，所以从句省略
- From both maturity to death, a wave is subject to the same laws as ~~(is) ~~any other 'living' thing. 主语太长可以倒装

※ For a time it assumes a miraculous individuality that, in the end, is reabsorbed into the great ocean of life.

『 **for a time** temporarily, for a certain period 暂时，一时，一度 』

- This method was used **for a time**, but it is no use now
- He was **for a time** a professor at Cambridge.
- Despite the long and complicated process, the project was successful **in the end**.尽管过程漫长而复杂，但该项目最终还是成功了。

『 **assume** v. (fml.) to start to have a particular quality or appearance = take on 开始拥有（某种特征或外表） 』

- These relationships **assume** great importance in times of crisis.

- In the story the god **assumes** the form of an eagle.

『**individuality** n. [U] the qualities that make sb. or sth. different from other things or people 个性，个人特征』

- We like our staff to show their **individuality** rather than wear a uniform.
- The plot is credible but the characters lack **individuality**.
- Each snowflake, briefly, **assumes a miraculous individuality** before melting.每一片雪花在融化之前，都会短暂地呈现出奇迹般的个体性。
- The child's imagination allowed him to **assume a miraculous individuality** in his play.孩子的想象力使他在玩耍中能够呈现出奇迹般的个体性。

『**reabsorb** v. to absorb again 再次吸收，使再次融入』

- The economy cannot **reabsorb** all of the refugees **into** the workforce.
- The kidneys **reabsorb** most of the water and nutrients back into the bloodstream. 肾脏会将大部分水分和养分重新吸收进血液中。
- After the rain, the water slowly **was reabsorbed into** the dry soil.雨后，水缓慢地被重新吸收到干燥的土壤中。

reabsorb ... into ... 将...重新吸收到...中；使...再次融入	After the merger, the smaller company was **reabsorbed into** the parent corporation. 合并后，这家小公司被重新并入母公司。

结论 (笔记最后修改于:2026-04-09 05:01:40) 3

※ The undulating waves of the open sea are generated by three natural causes: wind, earth movements or tremors, and the gravitational pull of the moon and the sun.

『**undulating** v. (fml.) to move or be shaped like waves that are rising and falling 波动，呈波浪状起伏』

- a field of wheat **undulating** in the breeze
- The artist perfectly captured the look of **the undulating waves** in the painting.这位艺术家在画中完美地捕捉了起伏的波浪的景象。
- We watched **the undulating waves** from the deck of the cruise ship.我们在游轮的甲板上观看着起伏的波浪。
- The surface of the lake began to **undulate** as the wind picked up. 随着风力增强，湖面开始泛起波纹。

undulating hills 起伏的山峦；连绵起伏的小山	The road wound through miles of **undulating hills** and green pastures. 道路穿过数英里连绵起伏的小山和绿色牧场。

『**be generated by** 由…产生/生成』

- The undulating waves of the open sea **are generated by three natural causes.** 开阔海域起伏的波浪由三种自然原因产生。
- Electricity **is generated by** various sources, including coal, wind, and solar. 电力由各种来源产生，包括煤炭、风能和太阳能。

『**tremor** n. 1. a small earthquake in which the ground shakes slightly 轻微地震』

- Minor earth movements or **tremors** are common in that fault zone. 轻微的地壳运动或震动在该断层带很常见。
- The **earth tremor** lasted only a few seconds, but it was enough to rattle the windows. 那次轻微地震仅持续了几秒钟，但足以震得窗户嘎嘎作响。

『**tremor** n. 2. a slight shaking movement in your body that you cannot control, especially because you are ill, weak, or upset 颤抖』

- There was a slight **tremor** in his voice.
- She felt a **tremor** of fear run through her.
- There was a slight **tremor** in her voice as she began to speak about the incident. 当她开始谈论那起事件时，声音略带颤抖。
- Parkinson's disease often causes an involuntary **tremor** in the hands. 帕金森病经常导致手部不由自主地颤抖。

『**gravitational** adj. related to or resulting from the force of gravity 重力的』

- The **gravitational** force between the Earth and the Moon keeps the Moon in orbit. 地球和月球之间的万有引力使月球保持在轨道上。
- The **gravitational pull of** Jupiter affects the orbits of many asteroids. 木星的引力影响着许多小行星的轨道。

gravitational field 重力场；引力场	An object within a planet's **gravitational field** will experience a force pulling it toward the center. 处于行星引力场内的物体会感受到一股将其拉向中心的力。
gravitational pull 引力作用；重力拉力	The **gravitational pull of** the sun is strong enough to hold the entire solar system together. 太阳的引力强大到足以将整个太阳系维系在一起。

※ Once waves have been generated, gravity is the force that drives them in a continual attempt to restore the ocean surface to a flat plain.

『**restore sth. to sth.** 恢复；使（某物）恢复到（某种状态/位置）』

- The committee is working hard to **restore** the old church **to** its original glory. 委员会正努力使这座古老的教堂恢复往日的辉煌。
- The government promises to **restore** the economy **to** full strength.
- She was hoping that the Mediterranean climate would **restore** her **to** full health.

- The engineers worked to **restore** the damaged area **to** a flat plain for construction.工程师
  们努力将受损区域恢复到平面以进行施工。



spill the beans

泄露秘密, 走漏风声

Idioms

请发送邮件至 hibenba@gmail.com 或直接访问官网 (**ncego.com**) 提交错误举报。

“一场孤独的修行，需要相信积累的力量。专注当下、珍惜时间、顶峰相见。”